Synthesis

Settings

Disconnections

Only show ones that fit the selected bond

FGI

R–OH → R–X

R–X → R–OH

Alkene → alkane

OXIDATION

R–OH → R=O

REDUCTION

R=O → R–OH

FGA

R–R → R=R

R–R → R–C=O

SPECIAL REACTIONS

Diels–Alder

Wittig

C–X DISCONNECTION

One group

Two group: 1,1 · 1,2 · 1,3

C–C DISCONNECTION

One group

Two group: 1,1 · 1,2 · 1,3

Aldol condensation

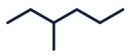
Claisen

More disconnections

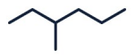
1 Retrosynthesis — 2 Forward synthesis

Restart New problem

TARGET



SYNTHESIS PLAN



INTERMEDIATES (TO DO)

None

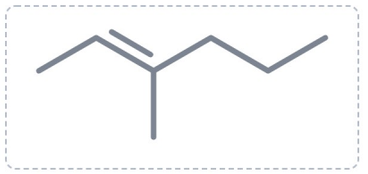
STARTING MATERIALS

None yet

TARGET / INTERMEDIATE MOLECULE



FGA: alkene



Pick a disconnection from the list, then click the bond to apply it to.

Apply disconnection

Clear

Unlimited practice for the organic chemistry problems that trip students up on exams.

PRACTICE

IUPAC naming

Synthesis

LEARN

Lectures

ACCOUNT

Sign in

Settings

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Synthesis

Settings

Disconnections

Only show ones that fit the selected bond

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One group

Two group: 1,1 · 1,2 · 1,3

C–C DISCONNECTION

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Two group: 1,1 · 1,2 · 1,3

Aldol condensation

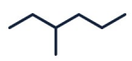
Claisen

More disconnections

1 Retrosynthesis — 2 Forward synthesis

Restart · New problem

TARGET



SYNTHESIS PLAN



INTERMEDIATES (TO DO)

None

STARTING MATERIALS

None yet

TARGET / INTERMEDIATE MOLECULE



Wittig

Option A



OR

Option B



Choose which way the bond breaks. Shaded molecules are starting materials, so they don't need to be made.

Use this option · Back

Unlimited practice for the organic chemistry problems that trip students up on exams.

PRACTICE

IUPAC naming

Synthesis

LEARN

Lectures

ACCOUNT

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Settings

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Synthesis

Settings

Disconnections

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R–X → R–OH

Alkene → alkane

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R–OH → R=O

REDUCTION

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FGA

R–R → R=R

R–R → R–C=O

SPECIAL REACTIONS

Diels–Alder

Wittig

C–X DISCONNECTION

One group

Two group: 1,1 · 1,2 · 1,3

C–C DISCONNECTION

One group

Two group: 1,1 · 1,2 · 1,3

Aldol condensation

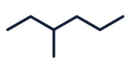
Claisen

More disconnections

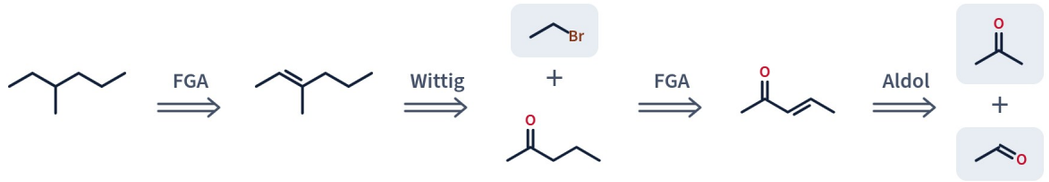
1 Retrosynthesis — 2 Forward synthesis

Restart · New problem

TARGET



SYNTHESIS PLAN



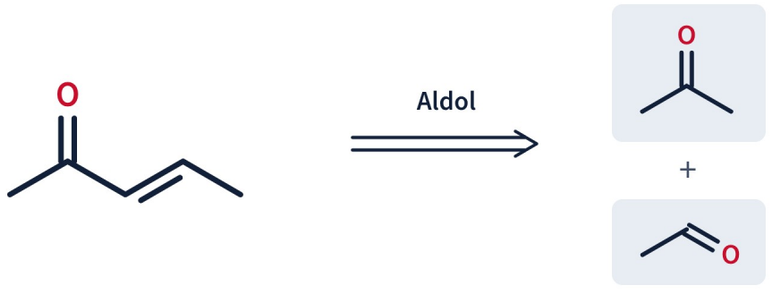
INTERMEDIATES (TO DO)

None

STARTING MATERIALS



TARGET / INTERMEDIATE MOLECULE



Retrosynthesis complete

Next, forward synthesis: match each step with the reagents or conditions that carry it out.

Every intermediate is now broken down into starting materials.

Start forward synthesis →

Review plan

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Synthesis

Settings

Reagents and conditions

Search reagents

CONDITIONS

Heat (Δ)

Dean–Stark trap (removes water)

H_3O^+

NaOH

REDUCING AGENTS

H_2, Pd/C

$NaBH_4$

$LiAlH_4$

OXIDIZING AGENTS

PCC

$KMnO_4$

PHOSPHORUS REAGENTS

PPh_3, then n-BuLi

ORGANOMETALLICS

Mg, ether

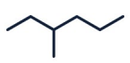
R–Li

Full reagent list

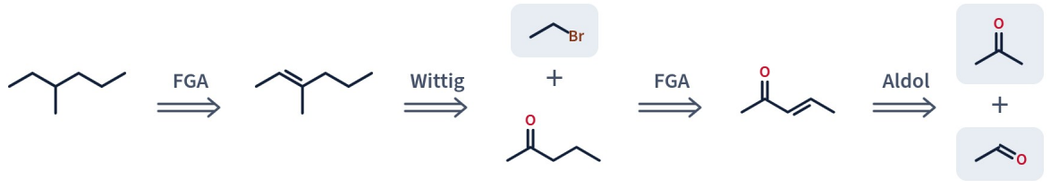
Retrosynthesis — 2 Forward synthesis

Restart · New problem

TARGET



SYNTHESIS PLAN



INTERMEDIATES (TO DO)

None

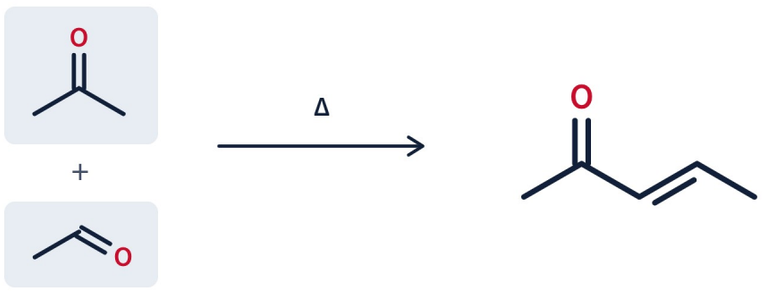
STARTING MATERIALS



FORWARD SYNTHESIS



CURRENT STEP



Note

This reaction can give the addition or the elimination product. To favor elimination, use heat (Δ).

No reagent to pick for this step: it uses heat, shown on the arrow.

Next step →

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Synthesis

Settings

Reagents and conditions

Search reagents

CONDITIONS

Heat (Δ)

Dean–Stark trap (removes water)

H_3O^+

NaOH

REDUCING AGENTS

H_2, Pd/C ×

$NaBH_4$

$LiAlH_4$

OXIDIZING AGENTS

PCC

$KMnO_4$

PHOSPHORUS REAGENTS

PPh_3, then n-BuLi

ORGANOMETALLICS

Mg, ether

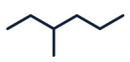
R–Li

Full reagent list

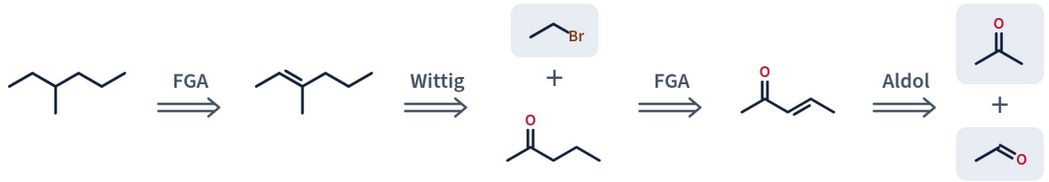
✓ Retrosynthesis — 2 Forward synthesis

Restart · New problem

TARGET



SYNTHESIS PLAN



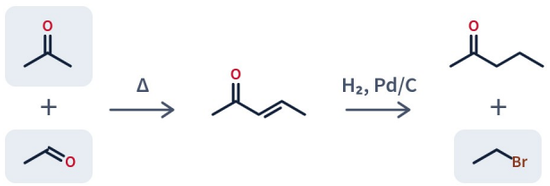
INTERMEDIATES (TO DO)

None

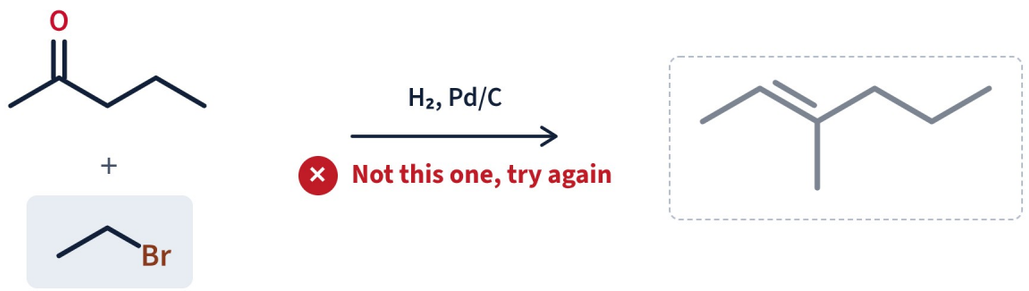
STARTING MATERIALS



FORWARD SYNTHESIS



CURRENT STEP



✕ Not this one, try again

Pick the reagent or conditions for this step from the list.

Next step →

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Synthesis

Settings

Reagents and conditions

Search reagents

CONDITIONS

Heat (Δ)

Dean–Stark trap (removes water)

H_3O^+

NaOH

REDUCING AGENTS

H_2, Pd/C

$NaBH_4$

$LiAlH_4$

OXIDIZING AGENTS

PCC

$KMnO_4$

PHOSPHORUS REAGENTS

PPh_3, then n-BuLi

ORGANOMETALLICS

Mg, ether

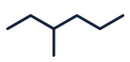
R–Li

Full reagent list

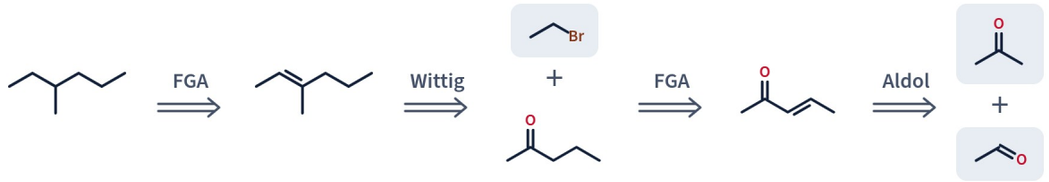
✓ Retrosynthesis — ✓ Forward synthesis

Restart · New problem

TARGET



SYNTHESIS PLAN



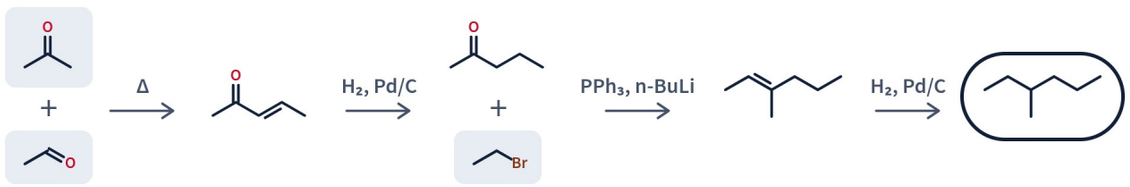
INTERMEDIATES (TO DO)

None

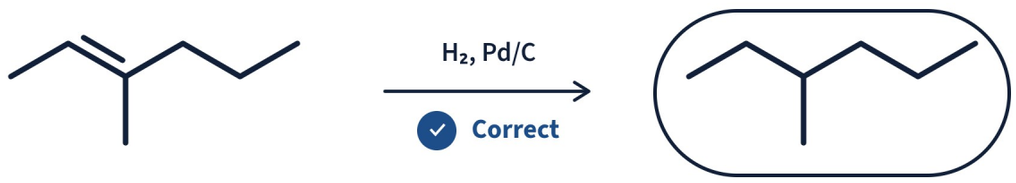
STARTING MATERIALS



FORWARD SYNTHESIS



CURRENT STEP



✓ Correct

Synthesis complete

You made the target from three starting materials in four steps.

The final product is circled at the end of the forward synthesis row.

New synthesis · Restart this synthesis

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Synthesis

Step 1 of 2 · **Retrosynthesis**

SYNTHESIS PLAN

Expand



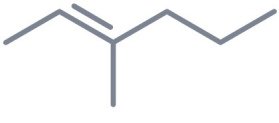
Intermediates: 0 Starting materials: 0

TARGET MOLECULE





FGA: alkene



Tap a bond, then pick a disconnection.

Disconnection

FGA: R–R → R=R

Clear

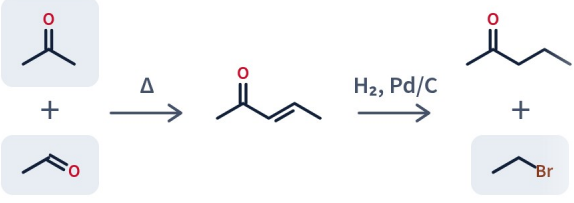
Apply

Synthesis

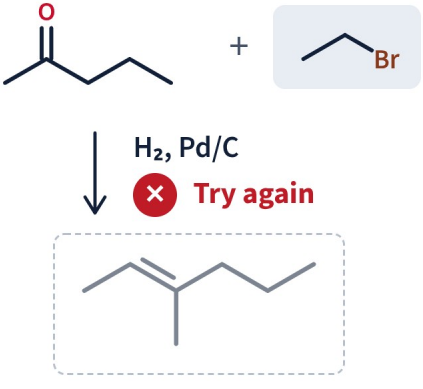
Step 2 of 2 · **Forward synthesis**

FORWARD SYNTHESIS

Expand



CURRENT STEP



Reagent

H_2, Pd/C

Next step →